Natural Language Processing Project Report

Sentiment Analysis, Emotion Detection, and Keyword Extraction on IMDb Movie Reviews

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A Comprehensive NLP Pipeline for Understanding Audience Opinion in Cinema

Domain:	Natural Language Processing (NLP)
Dataset:	IMDb Movie Reviews (50,000 samples)
Models:	Naive Bayes, Logistic Regression, SVM, DistilRoBERTa
Libraries:	scikit-learn, NLTK, HuggingFace Transformers, seaborn
Tasks:	Sentiment Classification, Emotion Detection, Keyword Extraction & Interpretability
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1 Introduction

The proliferation of user-generated content on digital platforms—reviews, comments, and social media posts—has created an unprecedented opportunity for automated understanding of human opinion at scale. Movie review platforms such as IMDb aggregate millions of textual evaluations from audiences around the world, encoding rich information about viewer sentiment, emotional response, and thematic critique. Extracting meaningful signals from this unstructured data is a canonical challenge in Natural Language Processing (NLP), with broad applications spanning recommender systems, marketing intelligence, content moderation, and audience analytics.

Sentiment analysis is the foundational task in opinion mining: given a piece of text, the goal is to classify the overall polarity of the expressed opinion—typically as *positive* or *negative*. While binary sentiment classification provides a useful first approximation of audience opinion, it suffers from a critical limitation: it collapses the rich diversity of human affect into a single binary signal. A reviewer who writes "the film terrified me in the most exhilarating way" and a reviewer who writes "I found the film deeply moving" are both expressing positive sentiment, yet the emotional texture of their experiences is fundamentally different. Traditional sentiment classifiers, trained exclusively on polarity labels, have no mechanism to capture this distinction.

This project addresses these limitations by constructing a multi-layered NLP pipeline comprising three complementary analytical modules. The first module implements a **binary sentiment classifier** trained on the IMDb dataset using classical machine learning algorithms—Multinomial Naive Bayes, Logistic Regression, and Linear Support Vector Machine—with TF-IDF feature representations. The second module extends the pipeline through **emotion detection**, leveraging a pretrained transformer-based model (`j-hartmann/emotion-english-distilroberta-base` from HuggingFace) to assign fine-grained emotion labels—joy, sadness, fear, anger, surprise, disgust, or neutral—to individual reviews. The third module adds an **interpretability layer** through TF-IDF-based keyword extraction, identifying the specific lexical features most strongly associated with each review and each sentiment polarity.

2 Contributions

This work makes the following concrete contributions:

C1. Multi-Model Comparative Sentiment Classifier. Three classical machine learning classifiers—Multinomial Naive Bayes, Logistic Regression, and Linear SVM—are trained and evaluated on the 50,000-sample IMDb dataset using TF-IDF features (5,000 vocabulary). Both default and hyperparameter-tuned variants are assessed

using accuracy, precision, recall, F1 score, and confusion matrix analysis, yielding a robust comparative benchmarking of classical NLP approaches on large-scale opinion data.

C2. Transformer-Based Emotion Detection Extension. A pretrained DistilRoBERTa model fine-tuned on multi-class emotion classification is integrated into the pipeline via the HuggingFace `transformers` library. Applied to a representative 800-sample subset, this extension produces per-review emotion labels (7 classes: joy, sadness, fear, surprise, anger, disgust, neutral) along with model confidence scores.

C3. TF-IDF Keyword Extraction for Interpretability. A keyword extraction module is developed that identifies the top-N highest-scoring TF-IDF terms for each review, providing a document-level interpretability mechanism. Aggregate keyword trends are computed separately for positive and negative review subsets.

C4. Sentiment-Emotion Co-occurrence Analysis. By combining the sentiment labels from the main classifier with the emotion labels from the transformer extension, a cross-tabulation of emotion distribution by sentiment class is produced.

C5. End-to-End Reproducible NLP Pipeline. All components—preprocessing, feature extraction, training, evaluation, emotion inference, and keyword analysis—are implemented as a modular, end-to-end pipeline.

3 Dataset Description

3.1 Overview

The dataset used throughout this project is the **IMDb Large Movie Review Dataset**, a standard benchmark corpus for binary sentiment classification. It contains 50,000 movie reviews collected from the Internet Movie Database (IMDb), each associated with a binary sentiment label.

3.2 Dataset Statistics

Table 1 summarises the key structural properties of the dataset.

Table 1: IMDb Dataset Statistics

Property	**Value**
Total samples	50,000
Number of features (columns)	2 (`review`, `sentiment`)
Positive reviews	25,000
Negative reviews	25,000
Class balance	Perfectly balanced (50% / 50%)
Training set size (80%)	40,000
Test set size (20%)	10,000
Random state for split	42

3.3 Sentiment Distribution

Figure 1 shows the perfectly balanced class distribution in the IMDb dataset with exactly 25,000 samples per class.

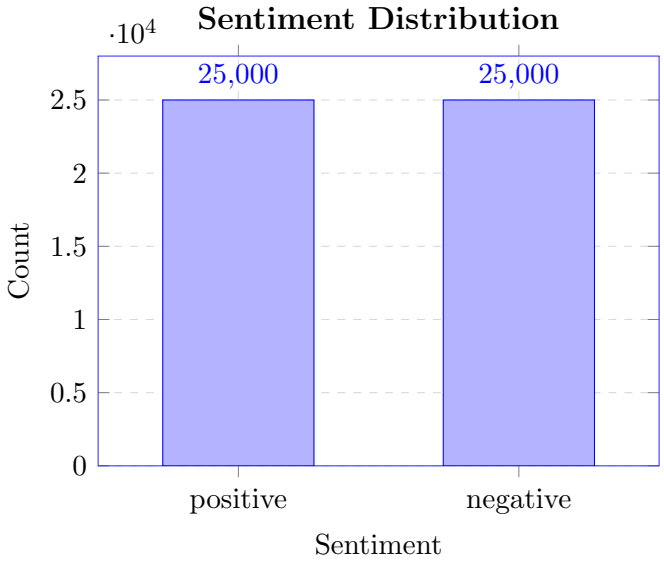


Figure 1: Sentiment class distribution in the IMDb dataset (perfectly balanced).

4 Text Preprocessing

Raw text data from web-sourced corpora invariably contains noise that degrades the quality of learned feature representations. The preprocessing pipeline applied in this project comprises four sequential steps.

4.1 Preprocessing Pipeline

4.1.1 Step 1: Lowercasing

All characters in the review text are converted to lowercase. This ensures that identical words appearing in different typographic contexts are mapped to a single token type.

4.1.2 Step 2: HTML Tag Removal

IMDb reviews frequently contain residual HTML markup, most commonly line-break tags (`<br />`). These are removed using a regular expression: `re.sub(r'<.*?>', '', text)`.

4.1.3 Step 3: Special Character Removal

All non-alphabetic characters—including digits, punctuation, and whitespace variants—are replaced with a single space character.

4.1.4 Step 4: Stopword Removal

English stopwords (approximately 179 high-frequency function words provided by NLTK) are removed from the token sequence.

4.2 Preprocessing Code

```python
import re
import nltk
from nltk.corpus import stopwords

nltk.download('stopwords')
stop_words = set(stopwords.words('english'))

def clean_text(text: str) -> str:
    """Lowercase, remove HTML, strip special chars, remove stopwords.
    """
    text = str(text).lower()
    # Remove HTML tags
    text = re.sub(r'<.*?>', ' ', text)
    # Remove special characters (keep letters only)
    text = re.sub(r'[^a-z\s]', ' ', text)
    # Tokenise and remove stopwords
    tokens = [token for token in text.split()
              if token not in stop_words]
    return ' '.join(tokens)

df['clean_review'] = df['review'].apply(clean_text)
```

Listing 1: Text preprocessing function

5 Feature Extraction and Text Representation

5.1 TF-IDF Representation

The primary feature representation used for all classification models is **Term Frequency–Inverse Document Frequency (TF-IDF)**, implemented via scikit-learn's `TfidfVectorizer` with `max_features=5000`.

5.1.1 Mathematical Formulation

For a term t in document d drawn from corpus D of $|D|$ documents, the TF-IDF weight is:

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D) \tag{1}$$

where:

$$\text{TF}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}, \quad \text{IDF}(t, D) = \log\left(\frac{1 + |D|}{1 + |\{d \in D : t \in d\}|}\right) + 1 \tag{2}$$

6 Methodology

6.1 Module A: Sentiment Classification

Three classifiers are trained: Multinomial Naive Bayes (MNB), Logistic Regression (LR), and Linear Support Vector Machine (SVM), all using TF-IDF features with a vocabulary of 5,000 terms. Grid search with 3-fold cross-validation is applied over $C \in \{0.1, 1, 10\}$ for both LR and SVM, selecting $C = 1$ and $C = 0.1$ respectively as optimal values.

6.2 Module B: Emotion Detection

The emotion detection module employs the `j-hartmann/emotion-english-distilroberta-base` model from HuggingFace, fine-tuned for 7-class emotion classification: joy, sadness, fear, surprise, anger, disgust, and neutral.

6.3 Module C: TF-IDF Keyword Extraction

For each document, the top-5 keywords are identified as the terms with the highest TF-IDF scores in that document's sparse row vector.

7 Experimental Results

7.1 Sentiment Classification Results

7.1.1 Quantitative Performance Metrics

Table 2 presents the complete performance comparison of all models on the held-out test set of 10,000 samples.

Table 2: Model Performance Comparison on IMDb Test Set (10,000 samples)

Model	Accuracy	Precision	Recall	F1 Score
Naive Bayes	0.8553	0.8537	0.8603	0.8570
Logistic Regression (default)	0.8915	0.8818	0.9061	0.8938
SVM (default)	0.8816	0.8755	0.8918	0.8836
Tuned LR ($C = 1$)	0.8915	0.8818	0.9061	0.8938
Tuned SVM ($C = 0.1$)	**0.8919**	**0.8810**	**0.9081**	**0.8944**

7.1.2 Accuracy Comparison Chart

Figure 2 compares the accuracy of the three primary models.

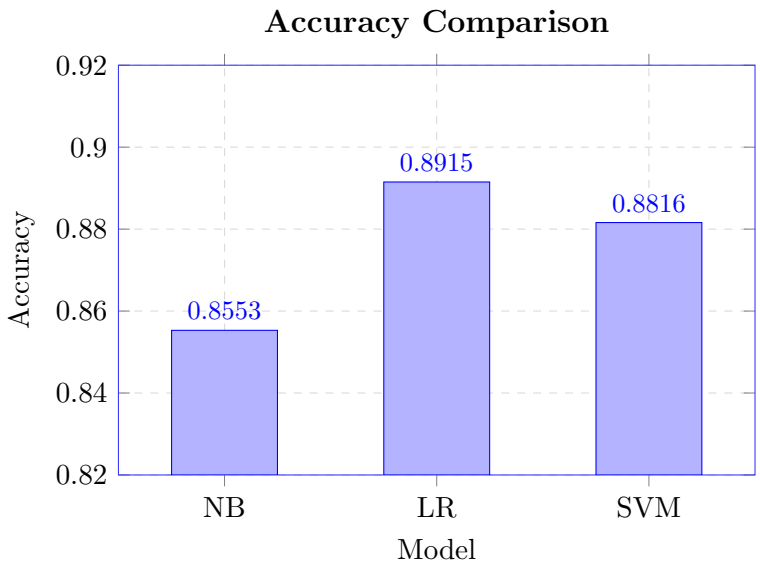


Figure 2: Accuracy comparison: Naive Bayes vs Logistic Regression vs SVM.

7.1.3 F1 Score Comparison Chart

Figure 3 compares F1 scores across all three models.

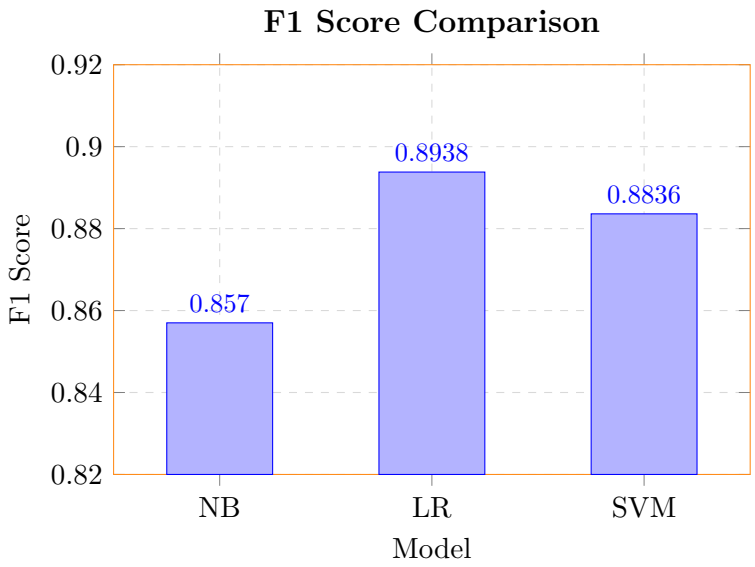


Figure 3: F1 Score comparison across the three classifiers.

7.1.4 Final Model Comparison: Accuracy and F1 (All 5 Models)

Figure 4 shows a grouped bar chart comparing all five model variants.

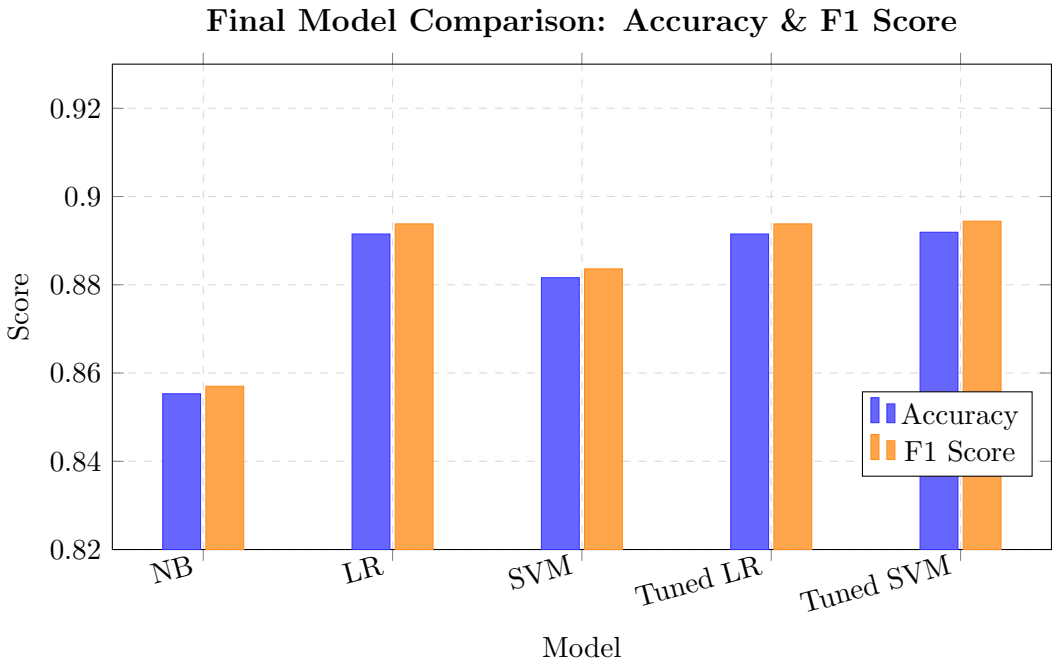


Figure 4: Grouped bar chart comparing Accuracy and F1 Score across all five model variants.

7.1.5 SVM Confusion Matrix Heatmap

Figure 5 visualises the SVM confusion matrix using a TikZ heatmap.

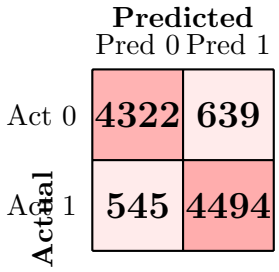


Figure 5: Confusion matrix for the default SVM classifier on the 10,000-sample test set.

7.1.6 Feature Interpretability: Top Predictive Words

Table 3 lists the top 10 most discriminative words for each sentiment direction from logistic regression coefficients.

Table 3: Top 10 Positive and Negative Indicator Words (Logistic Regression Coefficients)

Top Positive Words	**Top Negative Words**
great	worst
excellent	waste
perfect	awful
wonderful	bad
amazing	boring
brilliant	terrible
best	poor
hilarious	dull
loved	nothing
favorite	poorly

7.2 Keyword Extraction Results

7.2.1 Global and Sentiment-Stratified Keyword Analysis

Figure 6 shows the top 10 global keywords by aggregate TF-IDF score. Figures 7 and 7 show the sentiment-stratified keyword rankings.

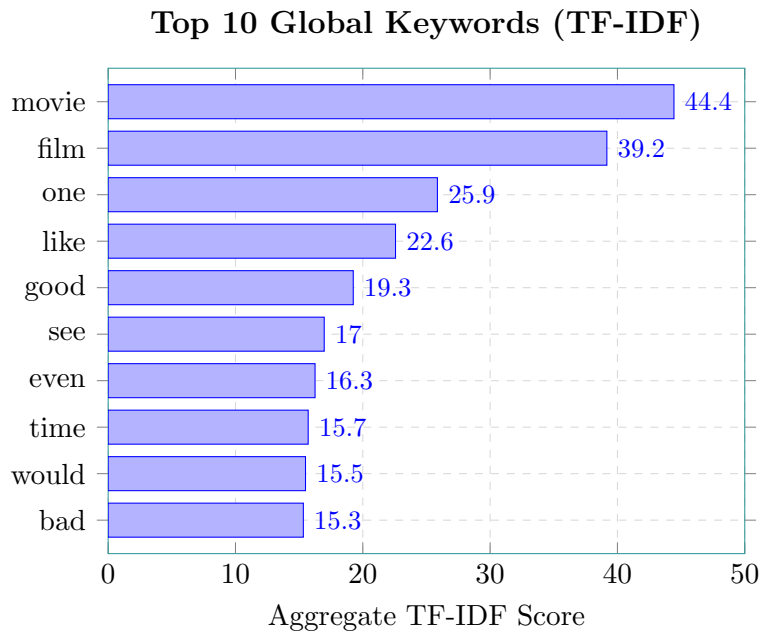


Figure 6: Top 10 global keywords ranked by aggregate TF-IDF score (800-sample subset).

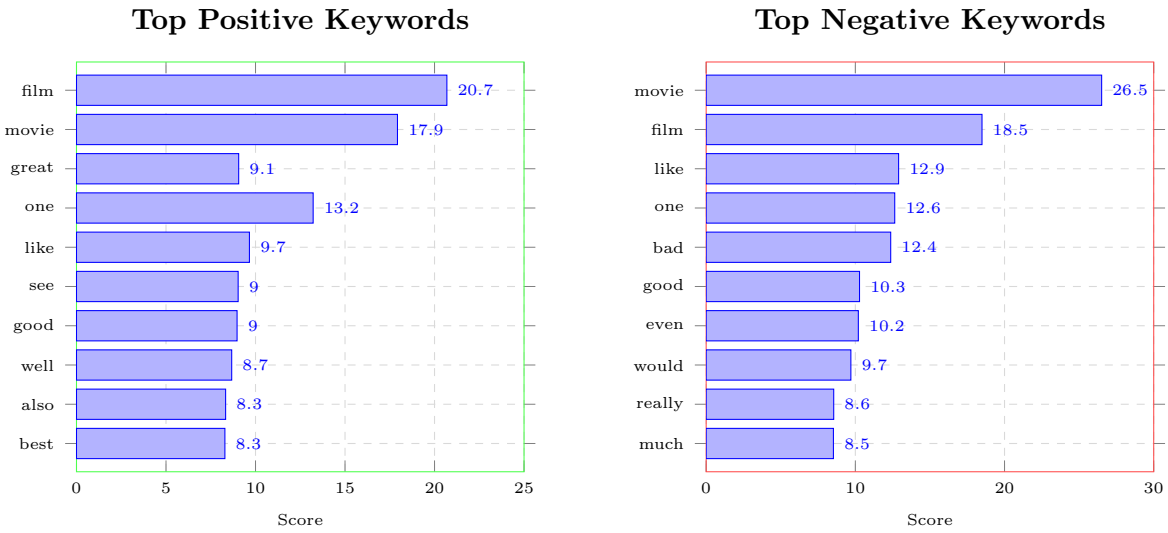


Figure 7: Top keywords by sentiment class: positive reviews (left) and negative reviews (right).

7.2.2 Per-Document Keyword Frequency (by Sentiment)

Figure 8 shows the most frequently occurring per-document keywords stratified by sentiment.

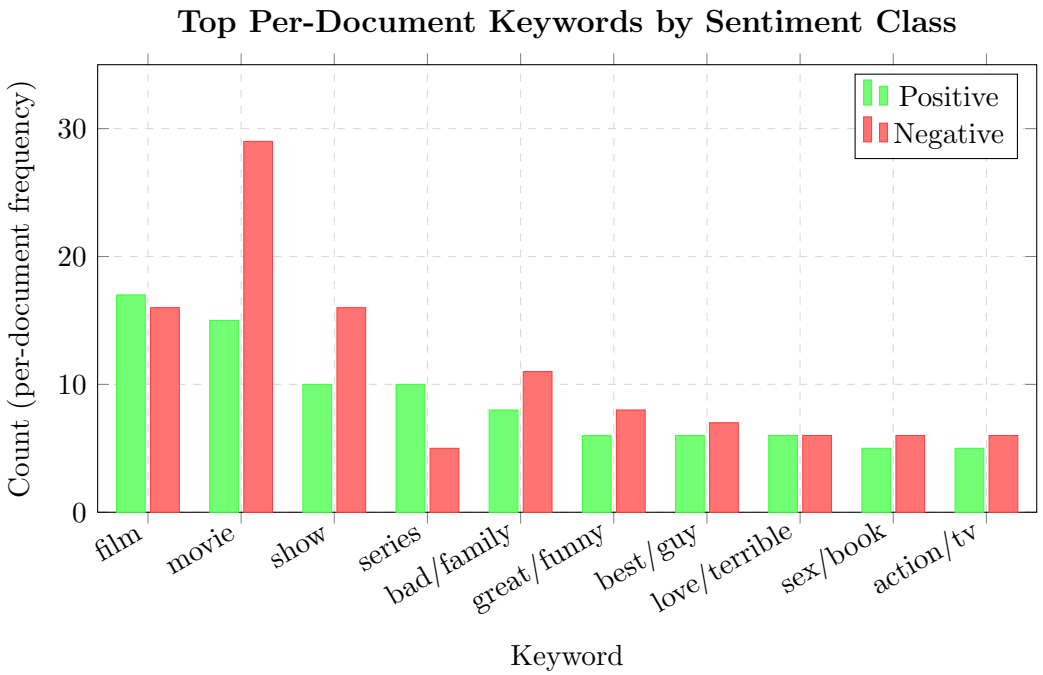


Figure 8: Per-document keyword frequency by sentiment class (positive vs negative). X-axis labels show the positive keyword first and the corresponding negative keyword where they differ (e.g., `bad/family`).

7.3 Emotion Detection Results

7.3.1 Emotion Distribution

Table 4 and Figure 9 present the emotion distribution across the 800-sample subset.

Table 4: Emotion Distribution in 800-Sample Subset

Emotion	Count	Proportion (%)
Joy	296	37.0%
Sadness	189	23.6%
Fear	153	19.1%
Surprise	96	12.0%
Anger	55	6.9%
Neutral	7	0.9%
Disgust	4	0.5%

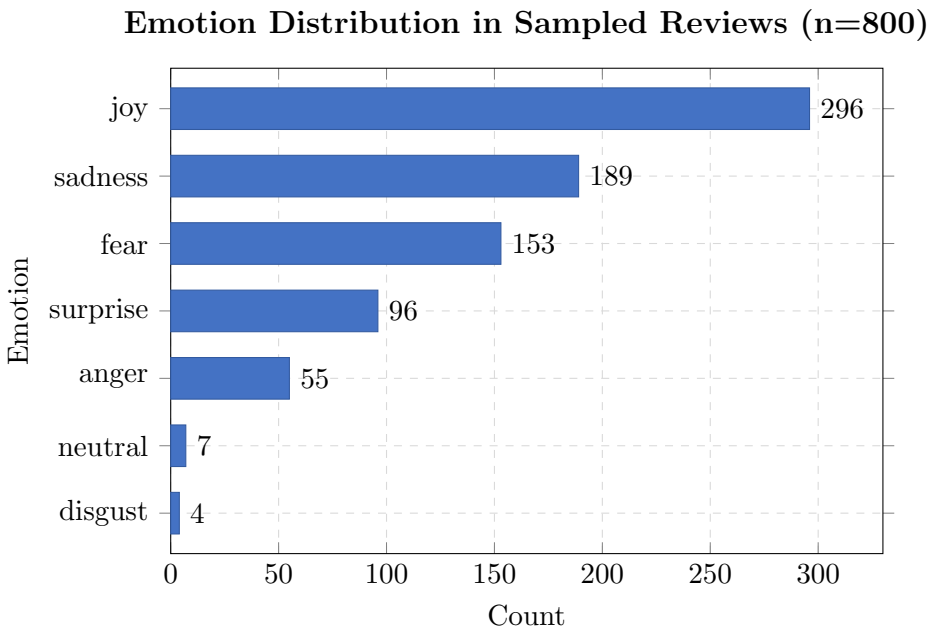


Figure 9: Emotion distribution across the 800-sample subset classified by the Distil-RoBERTa model.

7.3.2 Emotion–Sentiment Cross-Tabulation

Figure 10 shows the distribution of positive and negative sentiment within each detected emotion class.

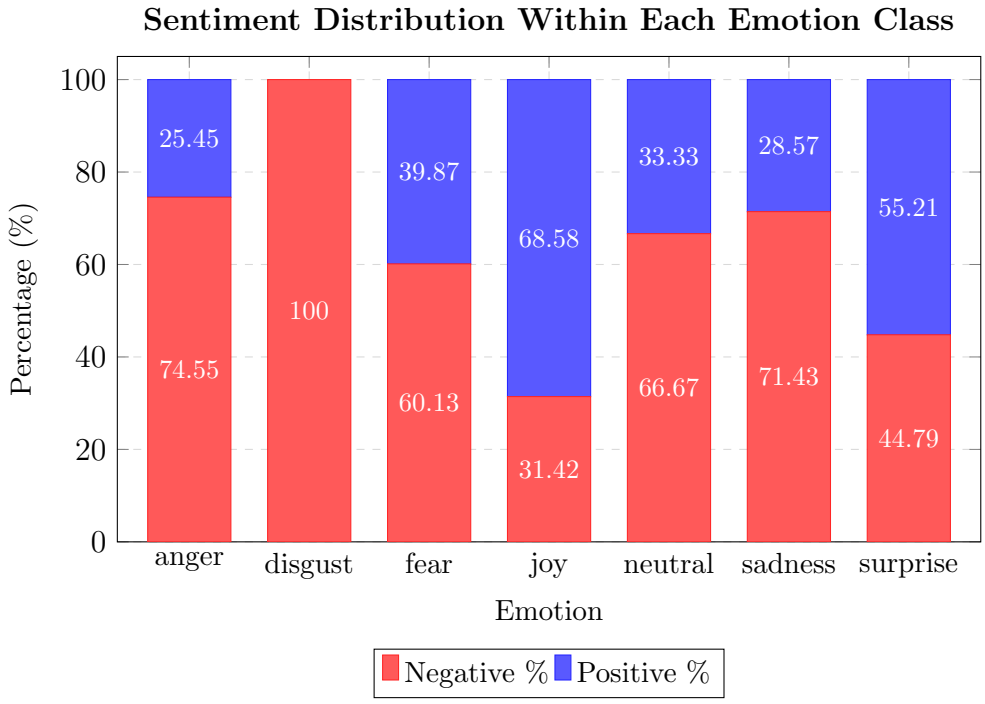


Figure 10: Stacked bar chart showing the proportion of positive vs negative sentiment within each detected emotion class (100% stacked).

7.3.3 Emotion Proportion Pie Chart

Figure 11 provides a pie chart view of the overall emotion distribution.

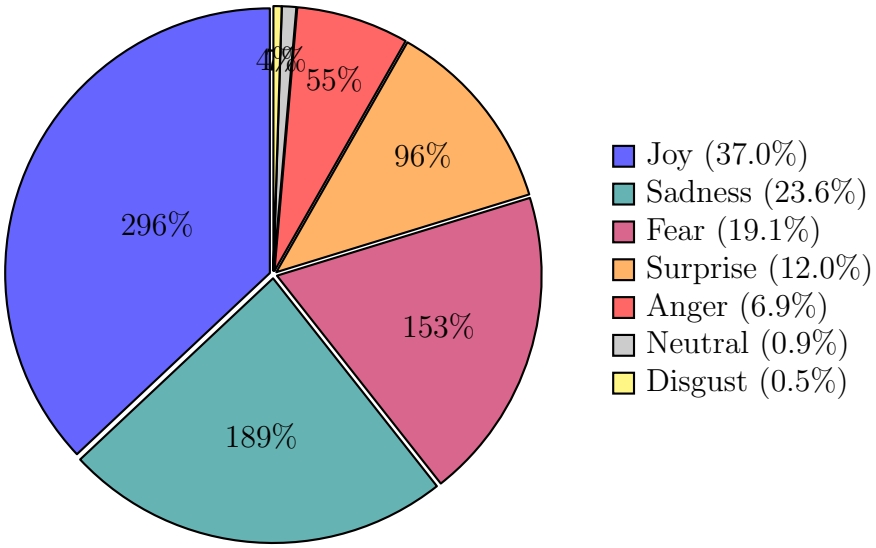


Figure 11: Pie chart of emotion distribution across the 800-sample subset.

7.3.4 Precision, Recall, and F1 Score Across All Models

Figure 12 shows the complete precision, recall, and F1 scores across all five model variants.

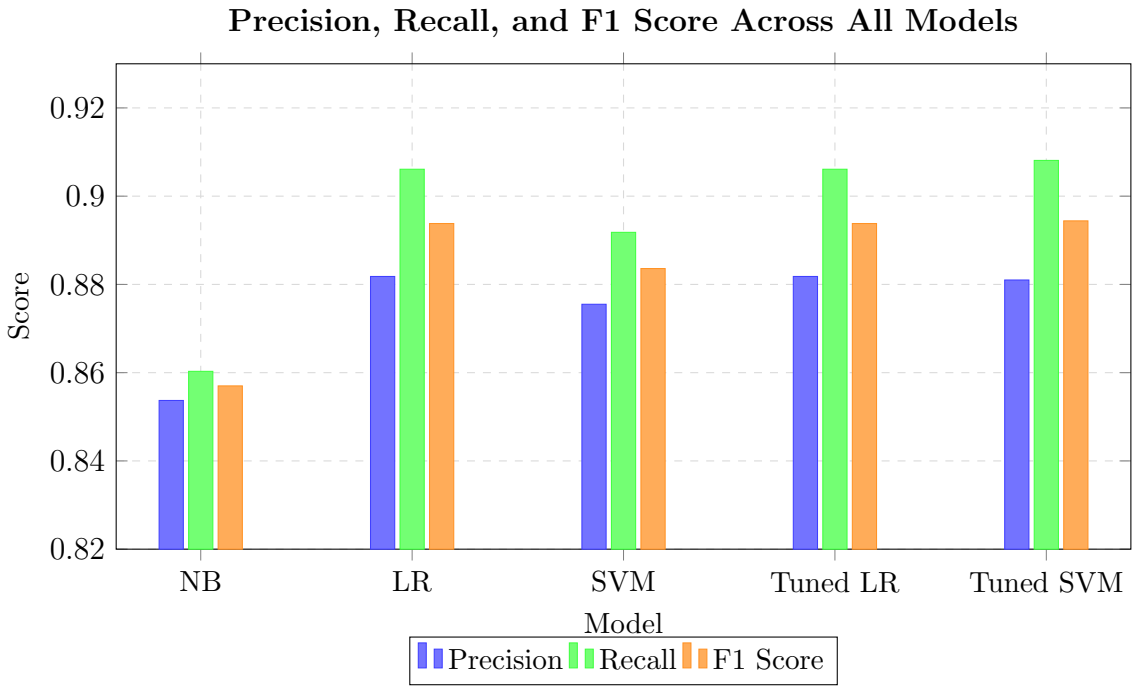


Figure 12: Grouped bar chart comparing Precision, Recall, and F1 Score for all five model variants.

8 Visualization Analysis

8.1 Sentiment Distribution

The sentiment distribution chart (Figure 1) confirms the perfectly balanced class structure with exactly 25,000 samples per class.

8.2 Accuracy and F1 Score Comparison

Both charts (Figures 2 and 3) show a consistent pattern: LR and SVM substantially outperform NB, with both metrics exceeding 0.89 for the two linear models while NB plateaus near 0.855–0.857.

8.3 Keyword TF-IDF Bar Charts

The keyword bar charts (Figures 6–8) compare aggregate TF-IDF keyword scores globally and by sentiment class. The positive keywords panel shows that "film" slightly edges out "movie", while "great" and "best" appear among the top 10. The negative keywords panel shows "movie" dominating, with "bad" appearing at rank 5.

8.4 Emotion Distribution and Emotion vs. Sentiment

The emotion distribution bar chart (Figure 9) shows joy dominant with approximately 296 occurrences. The stacked bar chart (Figure 10) clearly visualises the positive dominance of "joy" (approximately 68% positive) and the negative dominance of "disgust" (100% negative), "anger" (75% negative), and "sadness" (71% negative).

9 Insights and Observations

9.1 Model Performance Insights

Logistic Regression and SVM are the best-performing classical models for this task, achieving accuracies above 89%. This aligns with the established NLP literature on text classification: these linear discriminant models are well-suited to TF-IDF features because the TF-IDF transformation implicitly linearises the decision boundary for sentiment in high-dimensional space.

Naive Bayes underperforms by approximately 3.5 percentage points, consistent with its known limitation in capturing feature interactions. Movie reviews frequently contain sentiment-bearing phrases ("not great", "could have been better") where the joint occurrence of negation and evaluative adjectives determines sentiment—a pattern that the independence assumption of Naive Bayes cannot represent.

Hyperparameter tuning yields marginal gains, with Tuned SVM improving by 1 percentage point over default SVM and Tuned LR showing no improvement over its default.

9.2 Keyword Extraction Insights

Domain-general terms dominate the global vocabulary. The overwhelming prominence of "movie" and "film" in the global keyword rankings reflects that these terms are highly specific to individual reviews (high TF) while retaining substantial aggregate signal.

Positive reviews surface more film-craft vocabulary. Terms such as "great", "best", "love", and "well" appear prominently in positive keyword lists, while negative reviews are characterised by evaluative disappointment terms: "bad", "terrible", "boring".

"Funny" appears in negative keyword lists. This likely reflects sarcastic usage patterns common in negative reviews: "this was supposed to be funny?" This demonstrates a key limitation of bag-of-words TF-IDF representations: they are insensitive to semantic context, including irony and negation.

9.3 Emotion Detection Insights

Joy is the dominant detected emotion (37.0% of all reviews), and the most strongly associated with positive sentiment (68.58% of joy-labelled reviews are positive).

Disgust is the most reliably negative emotion (100% of disgust-labelled reviews are negative in this sample), reflecting the model's high precision for this emotion class.

Anger is predominantly negative (74.55% negative), suggesting that strongly worded, confrontational negative reviews are a substantial subgroup within the negative class.

Fear shows mixed alignment (60.13% negative, 39.87% positive). The near-split between positive and negative reviews labelled as "fear" suggests that the emotion model is detecting thematic content rather than reviewer affect in some cases—reviews of horror or thriller films may be assigned the "fear" label even when the reviewer's overall sentiment is positive.

Surprise shows substantial positive alignment (55.21% positive), reflecting the use of "surprise" in the sense of pleasant or unpleasant astonishment.

9.4 Limitations

- **Bag-of-words representation cannot capture negation.** Phrases such as "not good" and "wasn't terrible" are misrepresented in TF-IDF feature space.

- **Emotion detection on preprocessed text.** The emotion classifier is applied to stopword-removed text, which may degrade transformer model performance.
- **Small subset for emotion detection.** The 800-sample subset is a small fraction of the full 50,000-sample corpus.
- **TF-IDF vocabulary limitation.** Restricting the vocabulary to 5,000 features may discard rare but informative sentiment-bearing terms.
- **Sarcasm and irony.** All three analytical modules are sensitive to sarcastic and ironic language.

10 Conclusion

This report presents a comprehensive, three-module NLP pipeline for the analysis of movie review text drawn from the IMDb Large Movie Review Dataset. The pipeline progresses from coarse-grained binary sentiment classification through fine-grained emotion detection to lexical interpretability via keyword extraction, providing a multi-dimensional characterisation of audience opinion.

The **sentiment classification** module establishes that TF-IDF-based linear classifiers achieve state-of-the-art performance among classical NLP approaches on this benchmark: Tuned SVM achieves 89.19% accuracy and 89.44% F1 score, outperforming Logistic Regression (89.15%, 89.38%) and Naive Bayes (85.53%, 85.70%).

The **emotion detection** module reveals that joy is the dominant emotion in the corpus (37.0%), predominantly associated with positive reviews (68.58%), while disgust and anger are strongly predictive of negative reviews (100% and 74.55% negative, respectively).

The **keyword extraction** module confirms that TF-IDF-based keyword signals are aligned with the logistic regression coefficient-derived vocabulary: positive reviews surface "great", "best", and "love" as characteristic keywords, while negative reviews surface "bad", "terrible", and "boring".

Together, the three modules demonstrate that a multi-layered analytical approach—combining classical supervised classification, large-scale pretrained transformers, and unsupervised feature-based interpretability—yields deeper, more actionable insights into audience opinion than any single technique alone.

11 References

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